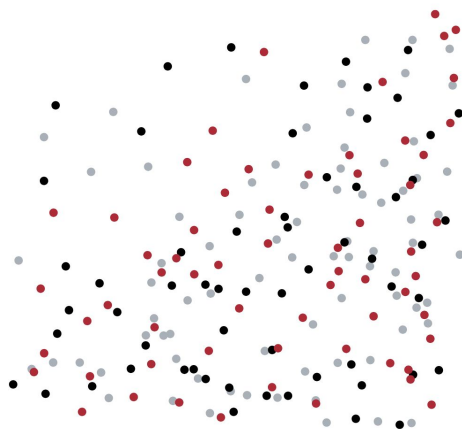


Environmental Data Analysis Report

Quantitative Data Analysis Report



Wildlife crossings | Island Fox Populations | Coral Bleaching | Ecosystem Carbon Storage

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Based on coursework from Foundations of Ecology and Environmental Biology (BB 1102); portfolio analysis expanded independently.

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INTRODUCTION

For this project, I analyzed four ecological and environmental datasets focused on island fox population change, wildlife crossing use, coral bleaching events, and ecosystem carbon storage. Each dataset addressed a real biological or environmental question, such as how animal populations change over time, how different structures affect wildlife movement, how bleaching events vary by severity, and how carbon storage differs across ecosystems.

The datasets included fox population numbers across different islands and years, percentages of wildlife, domestic animal, and human use on crossing structures, moderate and severe coral bleaching event counts over time, and carbon storage values in soil and vegetation across ecosystems. My aim was to organize the raw data, identify key variables, create visual summaries, calculate descriptive statistics, apply introductory inferential tests when appropriate, and explain what the results showed in a clear and evidence-based way.

DATA EXPLORATION

How The Data Was Collected

The datasets used in this report came from published ecological and environmental data sources. The Island Fox Populations dataset was modified from the 2019 Island Fox Status Update and showed population trends for six island fox subspecies over time. The Wildlife Crossings dataset came from Mysłajek et al. (2020) and recorded wildlife, domestic animal, and human use of different crossing structures. The Coral Bleaching dataset came from Our World in Data using data from Hughes (2018) and tracked moderate and severe coral bleaching events from 1980 to 2016. The Ecosystem Carbon Storage dataset was based on IPCC (2000) data and compared carbon stored in vegetation and soil across different ecosystems.

Features Identified for Analysis

The main features identified for analysis were year, island, island size, fox population, conservation status, crossing structure type, crossing percentages, bleaching severity, number of bleaching events, ecosystem type, carbon storage location, and tonnes of carbon stored. These features were selected because they allowed patterns, trends, and group differences to be compared using descriptive statistics and introductory inferential testing.

Ecosystem	Soil carbon storage (tonnes)	Vegetation carbon storage (tonnes)	Total carbon (tonnes)
Boreal forests	471	88	559
Temperate forests	100	59	159
Temperate grasslands	295	9	304
Tropical forests	216	212	428
Deserts and semideserts	191	8	199
Tundra	121	6	127
Wetlands	225	15	240
Tropical savannas	264	66	330
Cropland	128	2	130

Fig. 1. Raw carbon storage dataset by ecosystem

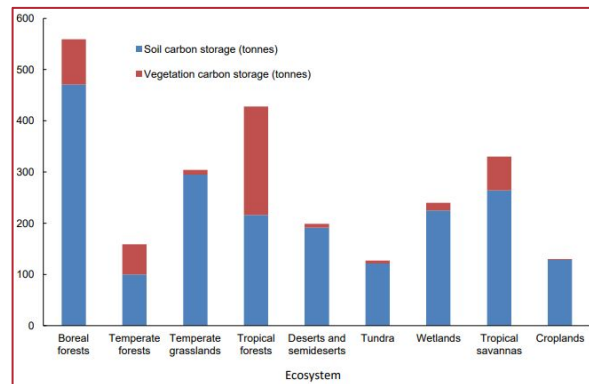


Fig. 2. Comparison of soil and vegetation carbon storage

METHODS

Analysis Techniques Used

Creating Visualizations: the first step was to create visualizations for each dataset. This made the patterns easier to see and helped compare trends, group differences, and relationships between variables.

Calculating Descriptive Statistics: after reviewing the visualizations, descriptive statistics were calculated. The mean was used to compare average values, and the standard deviation was used to measure how much variation existed within each group.

Selecting the Statistical Test: the statistical test was chosen based on the number of groups being compared. A t-test was used for two-group comparisons, while ANOVA was used for comparisons with three or more groups.

Running the Analysis: the main variables were inputted into the appropriate test to determine whether the observed differences between groups were meaningful.

Interpreting the Results: the p-value was used to interpret the test results. A p-value below 0.05 meant the null hypothesis was rejected, while a p-value above 0.05 meant there was not enough evidence to reject the null hypothesis.

WILDLIFE CROSSINGS - DATA EXPLORATION

Question: Does crossing structure type affect who uses the crossing structure?

Data used

The dataset listed crossing percentages for wildlife, domestic animals, and people across multiple crossing structure types. I reorganized the data so the structure type could be used to compare differences between groups.

Structure Type	Wildlife (%)	Domestic Animals (%)	People (%)
Overpass	75	5	20
Overpass	70	2	28
Overpass	55	5	40
Overpass	78	8	14
Overpass	85	5	10
Overpass	72	8	20
Large Underpass	40	25	35
Large Underpass	30	25	45
Large Underpass	40	20	40
Large Underpass	32	18	50
Large Underpass	38	25	37
Small Underpass	28	40	32
Small Underpass	25	10	65
Small Underpass	12	12	76
Small Underpass	22	12	66

Fig. 3. Wildlife crossing raw dataset

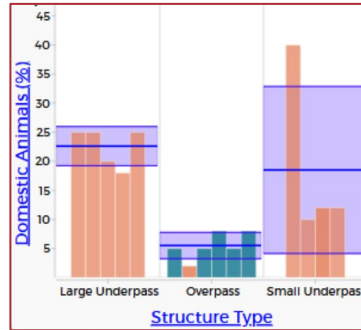


Fig. 4. Domestic animal crossing percentages by structure type

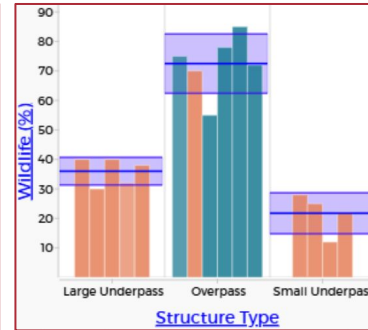


Fig. 5. Wildlife crossing percentages by structure type

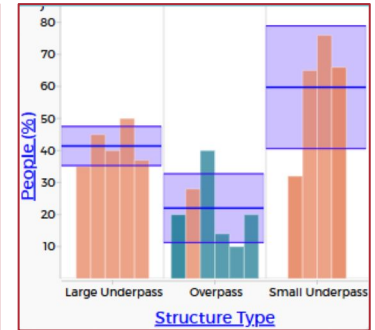


Fig. 6. People crossing percentages by structure type

Variables identified

Independent variable: structure type - large underpass, overpass, small underpass.

Dependent variables: percentage of domestic animal crossings, wildlife crossings, and people crossings.

Key comparison: which structure type had higher wildlife use versus higher human/domestic animal use.

What I checked before testing

I identified the explanatory variable and numerical response variables before choosing a statistical test. I used descriptive statistics, including mean and standard deviation, to compare the center and spread of each group. The statistical output was then used to determine whether the observed differences were meaningful.

WILDLIFE CROSSINGS - ANALYSIS AND RESULTS

Hypothesis tested

Null hypothesis 1: Crossing structure type has no effect on average wildlife crossing percentage.

Null hypothesis 2: There is no difference in average human crossing percentage between structure types.

Test used and why

I used ANOVA to compare crossing percentages across the three structure types: overpasses, large underpasses, and small underpasses. ANOVA was appropriate because the predictor variable had more than two groups and the response variable was numerical. I also used a t-test for selected two-group comparisons when comparing two structure types at a time.

Result

The ANOVA test showed a statistically significant difference between structure types, with $p < 0.05$. Since the p-value was less than 0.05, I rejected the null hypothesis. This means crossing structure type was associated with differences in crossing percentages.

Interpretation

The data suggest that structure type matters. Overpasses had the highest wildlife use, while small underpasses had higher human use. This means design type can influence whether crossings serve wildlife or are used more by people/domestic animals.

Uncertainty / limitation

The data show crossing percentages, but they do not explain every possible cause of crossing behavior. Location, surrounding habitat, time of day, species identity, traffic volume, and human activity could also affect how often crossings are used.

So what?

This analysis suggests that wildlife corridor design should consider structure type, not just the presence of a crossing. If future data tracked animal-vehicle collisions before and after installation, this type of analysis could help test whether certain crossing structures reduce collisions over time.

ISLAND FOX POPULATIONS - DATA EXPLORATION

Question: Are island fox population sizes different across islands over time?

Data used

The dataset tracked island fox population size from 1994 to 2018 across six islands. I graphed population trends and compared population values across islands.

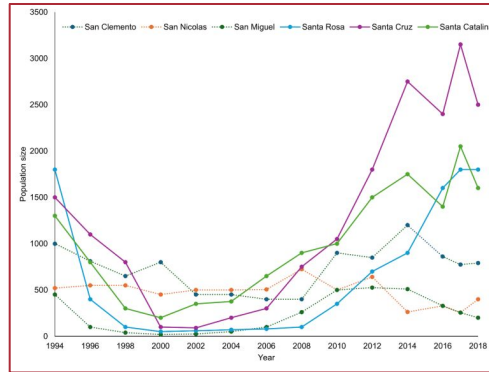


Fig. 7. Island fox population trends by island, 1994–2018

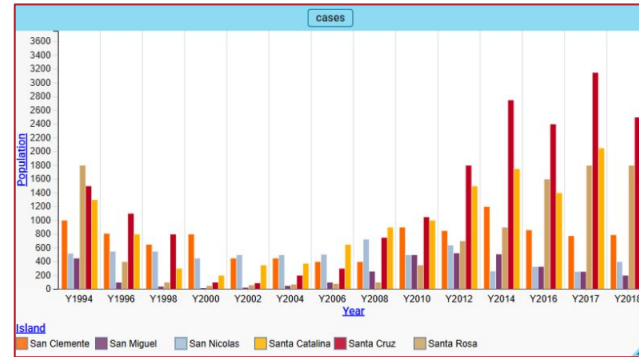


Fig. 8. Island fox population comparison by year and island

Variables identified

Independent variable: island, with year also used to understand population trend over time.

Dependent variable: fox population size.

Grouping factor: island subspecies / island location.

What I checked before testing

I first reviewed the population trends to see whether fox populations increased, decreased, or remained stable across islands. I then compared descriptive statistics, including mean and standard deviation, to understand which islands had higher average populations and which showed more variation over time. After comparing the groups, I used an ANOVA test to determine whether the differences in population size between islands were statistically meaningful.

This assignment included a reported ANOVA result with F and p values.

ISLAND FOX POPULATIONS - ANALYSIS AND RESULTS

Hypothesis tested

Null hypothesis: there is no difference in mean fox population between islands. Alternative hypothesis: there is a difference in mean fox population between islands.

Test used and why

I used a one-way ANOVA because the independent variable was island, which had multiple groups, and the dependent variable was numerical population size.

Result

The ANOVA test showed a significant difference in fox population between islands ($F = 5.837$, $p = 0.0001245$). Since $p < 0.05$, I rejected the null hypothesis.

Interpretation

Fox populations were not the same across islands. Santa Cruz had the highest and growing population pattern, while San Miguel was lower and more unstable. This suggests island conditions may strongly affect population recovery and stability.

Uncertainty / limitation

The analysis shows differences between islands, but it does not prove the exact cause. Island size, disease, predators, food availability, and human impact could all contribute.

So what?

A useful next question would be whether large islands support more stable fox recovery than small islands.

CORAL BLEACHING - DATA EXPLORATION

Question: Is there a significant difference between moderate and severe bleaching event counts across years?

Data used

The dataset tracked yearly counts of moderate and severe coral bleaching events from 1980 to 2016. I used the data to compare bleaching severity over time and examine whether severe bleaching events showed larger spikes in specific years.

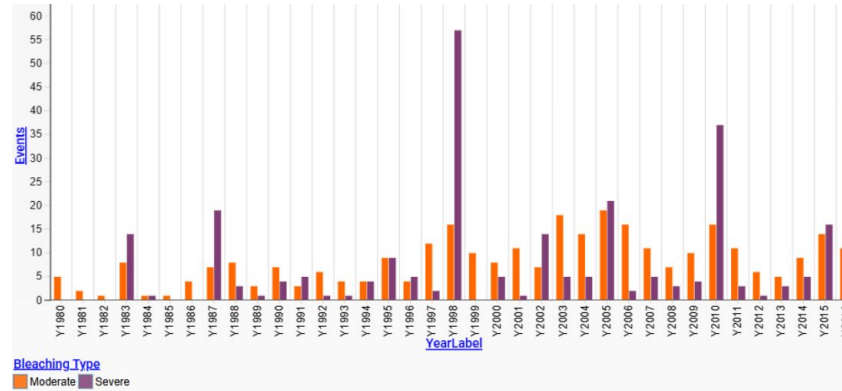


Fig. 9. Moderate and severe coral bleaching events, 1980–2016

Variables identified

Independent variables: year and bleaching type - moderate or severe.

Dependent variable: number of coral bleaching events.

Key comparison: moderate vs. severe event counts over the 1980-2016 time period.

What I checked before testing

I first reviewed the yearly bleaching pattern to see whether moderate and severe events followed similar trends over time. I then compared descriptive statistics, including mean and standard deviation, to understand the average number of events and the amount of variation in each bleaching type. Because the analysis compared two groups, moderate and severe bleaching, I used a two-sample t-test to determine whether the difference in mean event counts was statistically meaningful.

Analysis included descriptive statistics and a two-sample t-test comparing moderate and severe bleaching event counts.

CORAL BLEACHING - ANALYSIS AND RESULTS

Hypothesis tested

Null hypothesis: there is no difference in the mean number of bleaching events between moderate and severe bleaching types across years.

Test used and why

I used a two-sample t-test because the independent variable had two groups - moderate and severe - and the dependent variable was a numerical event count.

Result

The t-test result showed no statistically significant difference between the mean number of moderate and severe bleaching events, with $p > 0.05$. Therefore, I failed to reject the null hypothesis.

Interpretation

Even though severe bleaching had extreme spikes in years such as 1998, the mean difference between moderate and severe events was not statistically significant in this analysis.

Uncertainty / limitation

Averages can hide extreme years. Severe bleaching had a higher standard deviation, so the pattern may be about spikes and variability rather than only mean difference.

So what?

Policy insight: the increasing and extreme severe-event spikes suggest warming-related stress may make bleaching risk less predictable and more intense in certain years.

ECOSYSTEM CARBON STORAGE - DATA EXPLORATION

Question: Is carbon storage significantly different between soil and plant biomass across ecosystems?

Data used

The dataset listed carbon storage values for soil and vegetation across different terrestrial ecosystems. I used the data to compare how much carbon was stored below ground in soil versus above ground in plant biomass.

Ecosystem	Soil carbon storage (tonnes)	Vegetation carbon storage (tonnes)	Total carbon (tonnes)
Boreal forests	471	88	559
Temperate forests	100	59	159
Temperate grasslands	295	9	304
Tropical forests	216	212	428
Deserts and semideserts	191	8	199
Tundra	121	6	127
Wetlands	225	15	240
Tropical savannas	264	66	330
Cropland	128	2	130

Fig. 10. Raw carbon storage values by ecosystem

Variables identified

Independent variable: sequestration location - soil or plant.

Dependent variable: carbon stored in tonnes per hectare.

Ecosystem type was used to compare where carbon storage was highest.

What I checked before testing

I first compared the soil and vegetation carbon values across ecosystems to see which storage location held more carbon overall. I then used descriptive statistics, including mean and standard deviation, to compare the average carbon stored and the amount of variation in each group. Because the analysis compared two groups, soil and vegetation, I used a Welch two-sample t-test to determine whether the difference in mean carbon storage was statistically meaningful.

Analysis included descriptive statistics and a Welch two-sample t-test comparing soil and vegetation carbon storage.

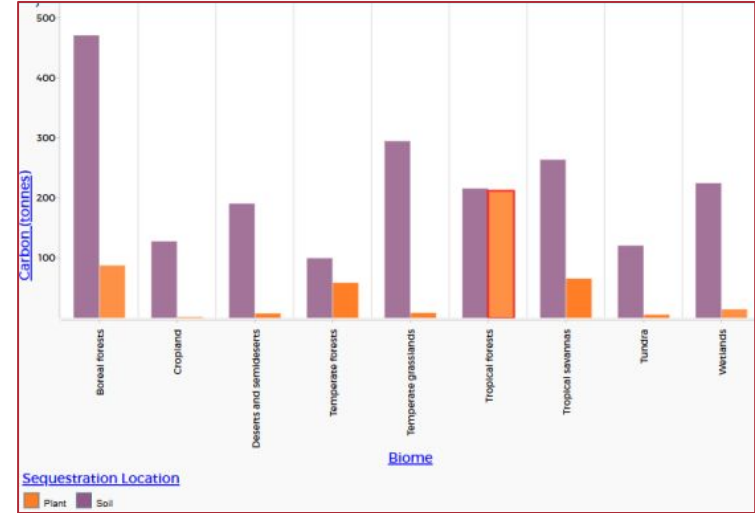


Fig. 11. Soil and vegetation carbon storage across ecosystems

ECOSYSTEM CARBON STORAGE - ANALYSIS AND RESULTS

Hypothesis tested

Null hypothesis: there is no difference in mean carbon storage between soil and plant locations across ecosystems.

Test used and why

I used a two-sample t-test because the analysis compared two groups - soil carbon and plant carbon - using continuous carbon storage values.

Result

The t-test showed a significant difference in carbon storage between soil and plant groups ($p = 0.001875$). Since $p < 0.05$, I rejected the null hypothesis.

Interpretation

Soil stored significantly more carbon than plant biomass across the ecosystems analyzed. The assignment mean values were about 223.4 tonnes for soil and 51.7 tonnes for plant carbon.

Uncertainty / limitation

This dataset compares broad ecosystem categories, not individual field measurements.
Human disturbance, land-use history, climate, and soil depth could change carbon values.

So what?

Conservation insight: protecting soils may be just as important as protecting visible plant biomass because soil can hold a large share of ecosystem carbon.

CONCLUSION

This report analyzed four ecological and environmental datasets to compare patterns, test group differences, and interpret statistical results. The analysis focused on wildlife crossing use, island fox population change, coral bleaching events, and ecosystem carbon storage. The results showed that crossing structure type affected wildlife crossing use, with $p < 0.05$. Island fox populations differed significantly across islands, with ANOVA showing $F = 5.837$ and $p = 0.0001245$. Coral bleaching showed no significant difference between mean moderate and severe event counts, but severe events had larger spikes in certain years. Ecosystem carbon storage showed that soil stored significantly more carbon than vegetation, with $p = 0.001875$. Overall, this work showed how descriptive statistics, t-tests, and ANOVA can be used to evaluate whether patterns in ecological data are statistically meaningful. It also showed the importance of explaining uncertainty and not overstating results beyond what the data can support.

REFERENCES

The datasets and original activity materials used in this report were provided through Foundations of Ecology and Environmental Biology (BB 1102). The analysis, written interpretation, organization, and portfolio formatting were completed independently.

Sources Used:

Mysłajek et al. (2020). Wildlife crossing structure data. European Journal of Wildlife Research.

Road Ecology Center. (2024). Roadkill, a Preventable Natural Disaster. University of California, Davis.

2019 Island Fox Status Update / Rachel Schleiger, LibreTexts. Island fox population data.

Hughes, T. P. (2018). Coral bleaching events dataset. Our World in Data.

IPCC. (2000). Ecosystem carbon storage data. Land Use, Land-Use Change and Forestry.